

Stimulation and recording chamber for the use of intact non-anesthetized earthworms

- CONSTRUCTION PLAN -

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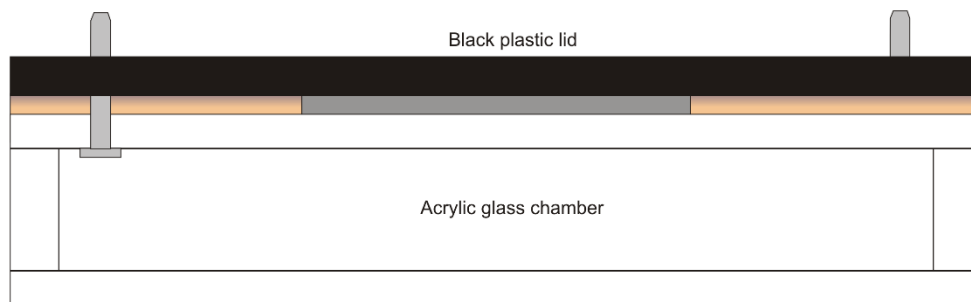
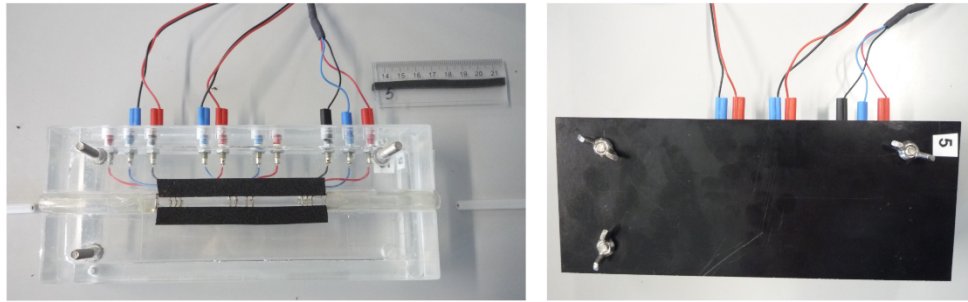
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Page 1: Overview of the earthworm chamber

Page 2: Chamber pieces, part 1

Page 3: Chamber pieces, part 2: recording platform

Overview of the earthworm chamber

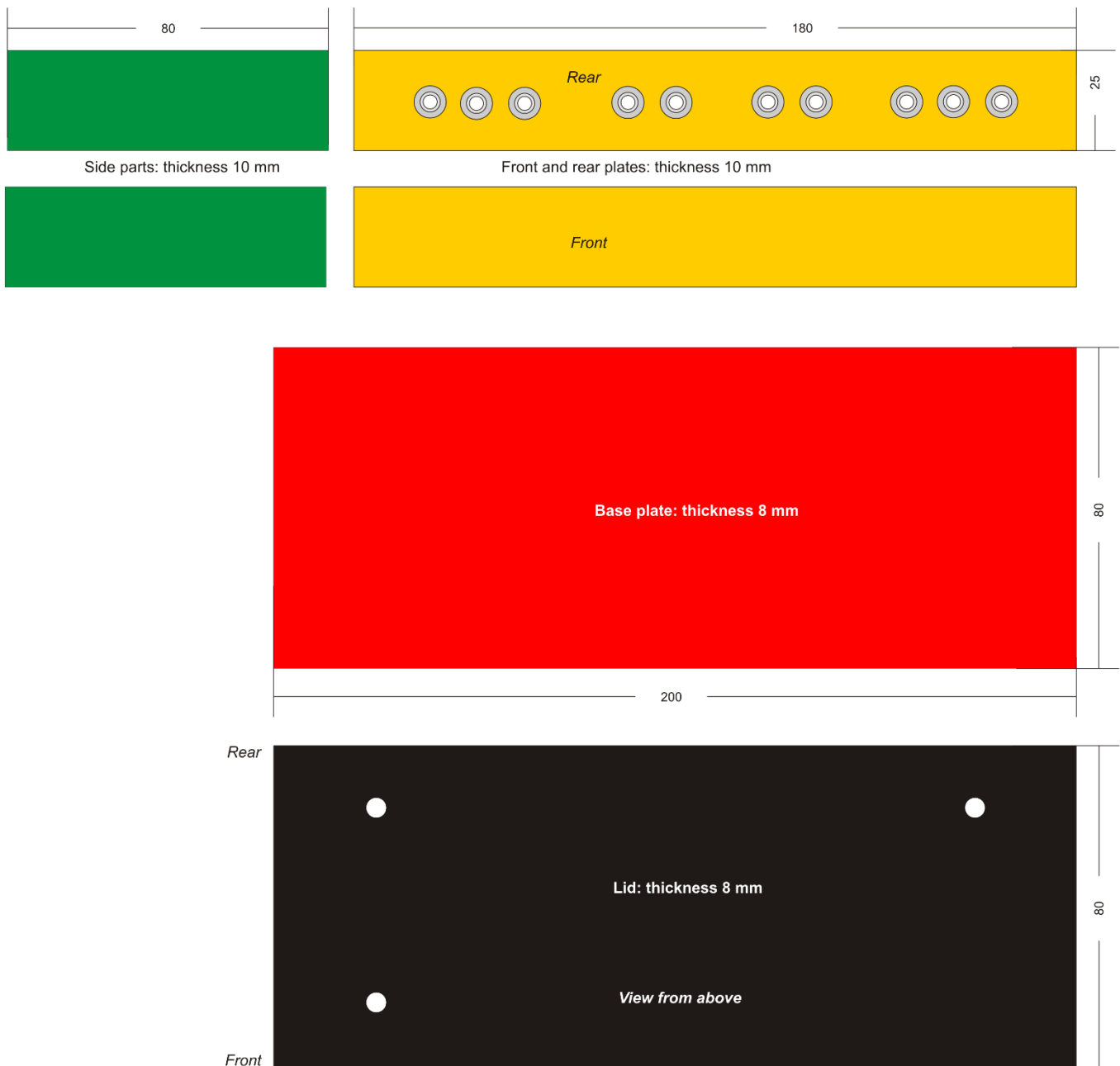


Color code of acrylic glass chamber pieces



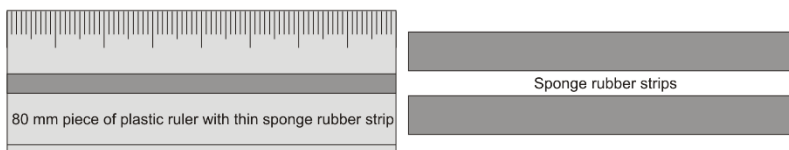
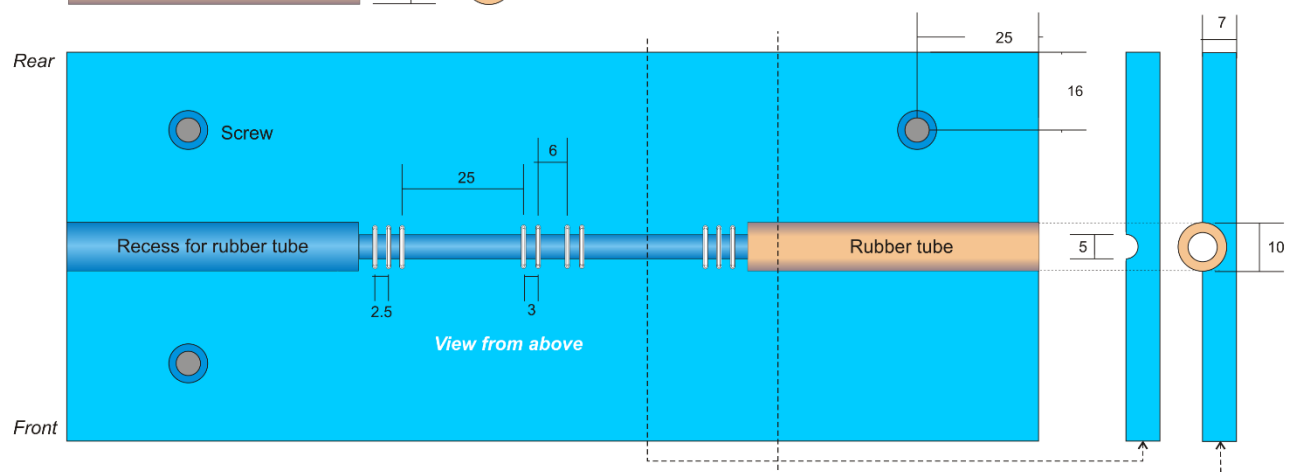
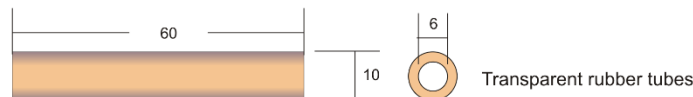
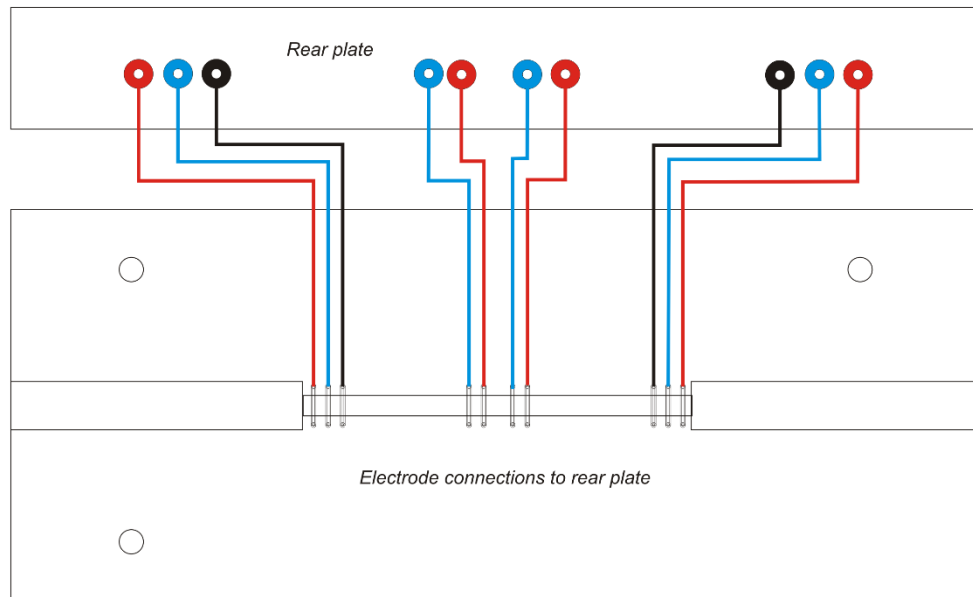
The photographs show the earthworm chamber viewed from above. Left side: open chamber (the recording platform of the acrylic glass chamber, including the lateral transparent rubber tubes and the central „halfpipe“ with the silver electrodes and the two elongated sponge rubber strips can be seen); the plugs (syringe needle covers) are not inserted; litz wires connect the electrodes with the pin jacks on the rear for the cable connections to the stimulation and recording equipment. Three screws are fixed in the platform from underneath, pointing outward. A piece of ruler, equipped with a thin sponge rubber strip, allows measurement of distances between electrodes and serves as a cover for the central duct. Right side: closed chamber (black plastic lid with three holes for the screws in the chamber is tightened with three wing nuts). The schematic drawings of the chamber show front views. Upper scheme: Between the black plastic lid and the acrylic glass chamber (with screws) the rubber tubes (brown) and the sponge rubber (grey) on the recording platform can be seen (see also page 3). Lower scheme: Color coding for the individual chamber pieces (side parts: green, front plate: yellow, base plate: red, recording platform: blue; see also page 2 and 3).

Chamber pieces, part 1 (without recording platform)



Individual chamber pieces are drawn with dimensions (in mm). The color coding refers to page 1. The rear plate has 10 holes for pin jacks (see page 3). The individual parts are glued together according to the scheme on page 1, except the base plate, which should be removable (screw connection) to allow easy access to the electrodes, litz wires and cable connections in the interior. The recording platform is shown on page 3.

Chamber pieces, part 2: recording platform



The recording platform harbors the central duct with the electrodes for the earthworm experiment. Pin jacks on the rear plate are connected to the individual silver electrodes on the recording platform: The lateral triplets are for stimulation (then a ground must be connected) or recording; two central pairs are routinely used for recording. The overall dimensions of the recording platform are identical to the ones of the base plate. Special features and dimensions are shown in the lower scheme. The rubber tube recesses and the central duct go half of their diameters deep into the platform. It is important to smooth the edges with sand paper after milling (especially in the central duct), because remaining sharp edges may injure the worm. Transparent rubber tubes are fixed (glued or sellotaped) in the lateral recesses. The sponge rubber strip on the ruler has to fit right between the two sponge rubber strips, which delineate the central duct. These sponge rubber strips normally cover the holes (10 on each side) for the silver wires (diameter: 0.5 mm), as can be seen on the photograph on page 1. Overall chamber dimensions and electrode distances are suggestions. However, the inner diameters of the rubber tubes (6 mm) and the central duct (5 mm) are critical.